

1. FT (Fourier Transform)
Continuous frequency analysis

Digital Discretization

2. FFT (Fast Fourier Transform)
Efficient overall frequency, but
lacks temporal localization

Need for Temporal Localization

3. STFT (Short-Time Fourier Transform)
Fixed sliding window (Heisenberg-Gabor
limit)

Need for Multi-Resolution

4. CWT (Continuous Wavelet Transform)
Variable window, but computationally heavy
& redundant

Pyramidal Sub-Band Coding

5. DWT (Discrete Wavelet Transform)
Fast $O(N)$ execution, filter banks,
minimized redundancy